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# =====
# IYR COMPLETE DATA SCIENCE WORKFLOW
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error

# -----
# 1. LOAD DATA
# -----

df_iyr = pd.read_csv("IYR.csv")

df_iyr["Date"] = pd.to_datetime(df_iyr["Date"])

df_iyr = (
    df_iyr
    .sort_values("Date")
    .drop_duplicates(subset=["Date"])
    .reset_index(drop=True)
)

print("IYR data loaded successfully.")
print("Shape:", df_iyr.shape)

display(df_iyr.head())

# -----
# 2. DAILY RETURN
# -----
```

```
df_iyr["Daily_Return"] = df_iyr["Close"].pct_change()

# -----
# 3. MOVING AVERAGES
# -----

df_iyr["MA_10"] = df_iyr["Close"].rolling(10).mean()
df_iyr["MA_20"] = df_iyr["Close"].rolling(20).mean()
df_iyr["MA_50"] = df_iyr["Close"].rolling(50).mean()

# -----
# 4. VOLATILITY
# -----

df_iyr["Volatility_20"] = (
    df_iyr["Daily_Return"].rolling(20).std()
)

# -----
# 5. RSI
# -----

delta = df_iyr["Close"].diff()

gain = delta.clip(lower=0)
loss = -delta.clip(upper=0)

avg_gain = gain.rolling(14).mean()
avg_loss = loss.rolling(14).mean()

rs = avg_gain / avg_loss

df_iyr["RSI_14"] = 100 - (100 / (1 + rs))
```

```
# -----  
# 6. MACD  
# -----  
  
ema_12 = df_iyr["Close"].ewm(  
    span=12,  
    adjust=False  
) .mean()  
  
ema_26 = df_iyr["Close"].ewm(  
    span=26,  
    adjust=False  
) .mean()  
  
df_iyr["MACD"] = ema_12 - ema_26  
  
df_iyr["MACD_Signal"] = (  
    df_iyr["MACD"]  
    .ewm(span=9, adjust=False)  
    .mean()  
)  
  
# -----  
# 7. SAVE FIGURES  
# -----  
  
# Closing Price  
plt.figure(figsize=(12, 6))  
plt.plot(df_iyr["Date"], df_iyr["Close"])  
plt.title("IYR Closing Price")  
plt.xlabel("Date")  
plt.ylabel("Closing Price")  
plt.xticks(rotation=45)  
plt.tight_layout()  
  
file_path = "IYR_Closing_Price.png"
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```
plt.savefig(file_path, dpi=300, bbox_inches="tight")
plt.show()
files.download(file_path)

# Daily Returns
plt.figure(figsize=(12, 6))
plt.plot(df_iyr["Date"], df_iyr["Daily_Return"])
plt.title("IYR Daily Returns")
plt.xlabel("Date")
plt.ylabel("Daily Return")
plt.xticks(rotation=45)
plt.tight_layout()

file_path = "IYR_Daily_Returns.png"
plt.savefig(file_path, dpi=300, bbox_inches="tight")
plt.show()
files.download(file_path)

# Moving Averages
plt.figure(figsize=(12, 6))
plt.plot(df_iyr["Date"], df_iyr["Close"], label="Close")
plt.plot(df_iyr["Date"], df_iyr["MA_10"], label="MA 10")
plt.plot(df_iyr["Date"], df_iyr["MA_20"], label="MA 20")
plt.plot(df_iyr["Date"], df_iyr["MA_50"], label="MA 50")
plt.title("IYR Closing Price and Moving Averages")
plt.xlabel("Date")
plt.ylabel("Price")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

file_path = "IYR_Moving_Averages.png"
plt.savefig(file_path, dpi=300, bbox_inches="tight")
plt.show()
files.download(file_path)
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```
# Volatility
plt.figure(figsize=(12, 6))
plt.plot(
    df_iyr["Date"],
    df_iyr["Volatility_20"]
)
plt.title("IYR 20-Day Rolling Volatility")
plt.xlabel("Date")
plt.ylabel("Volatility")
plt.xticks(rotation=45)
plt.tight_layout()

file_path = "IYR_Volatility_20.png"
plt.savefig(file_path, dpi=300, bbox_inches="tight")
plt.show()
files.download(file_path)

# RSI
plt.figure(figsize=(12, 6))
plt.plot(
    df_iyr["Date"],
    df_iyr["RSI_14"]
)
plt.axhline(70, linestyle="--")
plt.axhline(30, linestyle="--")
plt.title("IYR 14-Day RSI")
plt.xlabel("Date")
plt.ylabel("RSI")
plt.xticks(rotation=45)
plt.tight_layout()

file_path = "IYR_RSI_14.png"
plt.savefig(file_path, dpi=300, bbox_inches="tight")
plt.show()
files.download(file_path)
```

```
# MACD
plt.figure(figsize=(12, 6))
plt.plot(
    df_iyr["Date"],
    df_iyr["MACD"],
    label="MACD"
)
plt.plot(
    df_iyr["Date"],
    df_iyr["MACD_Signal"],
    label="MACD Signal"
)
plt.axhline(0, linestyle="--")
plt.title("IYR MACD and MACD Signal")
plt.xlabel("Date")
plt.ylabel("MACD")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

file_path = "IYR_MACD.png"
plt.savefig(file_path, dpi=300, bbox_inches="tight")
plt.show()
files.download(file_path)

# -----
# 8. PROCESSED DATASET
# -----

feature_columns = [
    "Date",
    "Open",
    "High",
    "Low",
    "Close",
    "Adj Close",
```

```
    "Volume",
    "Daily_Return",
    "MA_10",
    "MA_20",
    "MA_50",
    "Volatility_20",
    "RSI_14",
    "MACD",
    "MACD_Signal"
]

df_iyr_features = (
    df_iyr[feature_columns]
    .dropna()
    .reset_index(drop=True)
)

print("Processed IYR shape:", df_iyr_features.shape)

df_iyr_features.to_csv(
    "IYR_features.csv",
    index=False
)

files.download("IYR_features.csv")

# -----
# 9. TARGET
# -----

df_ml_iyr = df_iyr_features.copy()

df_ml_iyr["Target"] = (
    df_ml_iyr["Close"].shift(-1)
)

df_ml_iyr = (
```



```
df_ml_iyr
.dropna()
.reset_index(drop=True)
)

# -----
# 10. FEATURES FOR MODEL
# -----

model_features = [
    "Close",
    "Daily_Return",
    "MA_10",
    "MA_20",
    "MA_50",
    "Volatility_20",
    "RSI_14",
    "MACD",
    "MACD_Signal"
]

X = df_ml_iyr[model_features]
y = df_ml_iyr["Target"]

# -----
# 11. TIME-SERIES TRAIN/TEST SPLIT
# -----

split_index = int(len(df_ml_iyr) * 0.80)

X_train = X.iloc[:split_index]
X_test = X.iloc[split_index:]

y_train = y.iloc[:split_index]
y_test = y.iloc[split_index:]
```

```
print("Training samples:", len(X_train))
print("Testing samples:", len(X_test))

# -----
# 12. SCALING
# -----

scaler_iyr = StandardScaler()

X_train_scaled = scaler_iyr.fit_transform(X_train)
X_test_scaled = scaler_iyr.transform(X_test)

# -----
# 13. TRAIN MODEL
# -----

model_iyr = LinearRegression()

model_iyr.fit(
    X_train_scaled,
    y_train
)

print("IYR model training completed successfully.")

# -----
# 14. PREDICTION
# -----

y_pred_iyr = model_iyr.predict(
    X_test_scaled
)

print("IYR prediction completed successfully.")
```

```
# -----  
# 15. EVALUATION  
# -----  
  
actual_iyr = np.array(y_test)  
predicted_iyr = np.array(y_pred_iyr)  
  
mae_iyr = mean_absolute_error(  
    actual_iyr,  
    predicted_iyr  
)  
  
mse_iyr = mean_squared_error(  
    actual_iyr,  
    predicted_iyr  
)  
  
rmse_iyr = np.sqrt(mse_iyr)  
  
r2_iyr = r2_score(  
    actual_iyr,  
    predicted_iyr  
)  
  
print("\n")  
print("=" * 50)  
print("IYR MODEL EVALUATION RESULTS")  
print("=" * 50)  
print(f"MAE   : {mae_iyr:.4f}")  
print(f"MSE   : {mse_iyr:.4f}")  
print(f"RMSE  : {rmse_iyr:.4f}")  
print(f"R²    : {r2_iyr:.4f}")  
print("=" * 50)  
  
# -----  
# 16. ACTUAL VS PREDICTED
```

```
# 10. ACTUAL VS PREDICTED
# -----

test_dates_iyr = (
    df_ml_iyr["Date"]
    .iloc[split_index:]
    .reset_index(drop=True)
)

plt.figure(figsize=(12, 6))

plt.plot(
    test_dates_iyr,
    actual_iyr,
    label="Actual Price"
)

plt.plot(
    test_dates_iyr,
    predicted_iyr,
    label="Predicted Price"
)

plt.title("IYR Actual vs Predicted Closing Price")
plt.xlabel("Date")
plt.ylabel("Closing Price")
plt.legend()
plt.xticks(rotation=45)

plt.tight_layout()

file_path = "IYR_Actual_vs_Predicted.png"

plt.savefig(
    file_path,
    dpi=300,
    bbox_inches="tight"
)
```

```
plt.show()

files.download(file_path)

# -----
# 17. SAVE EVALUATION
# -----

evaluation_iyr = pd.DataFrame({
    "Metric": [
        "MAE",
        "MSE",
        "RMSE",
        "R2"
    ],
    "Value": [
        mae_iyr,
        mse_iyr,
        rmse_iyr,
        r2_iyr
    ]
})

evaluation_iyr.to_csv(
    "IYR_Evaluation_Metrics.csv",
    index=False
)

files.download(
    "IYR_Evaluation_Metrics.csv"
)

# -----
# 18. SAVE PREDICTIONS
# -----
```

```
prediction_iyr = pd.DataFrame({
    "Date": test_dates_iyr,
    "Actual": actual_iyr,
    "Predicted": predicted_iyr
})

prediction_iyr["Error"] = (
    prediction_iyr["Actual"] -
    prediction_iyr["Predicted"]
)

prediction_iyr.to_csv(
    "IYR_Prediction_Results.csv",
    index=False
)

files.download(
    "IYR_Prediction_Results.csv"
)

print("\nIYR COMPLETE WORKFLOW FINISHED SUCCESSFULLY.")
```

IYR data loaded successfully.
Shape: (3774, 7)

	Date	Open	High	Low	Close	Adj Close	Volume	
0	2010-01-04	46.450001	46.660000	45.529999	45.810001	26.637310	16952700	
1	2010-01-05	45.810001	45.959999	45.290001	45.919998	26.701263	14523000	
2	2010-01-06	45.939999	46.230000	45.709999	45.900002	26.689632	14859100	
3	2010-01-07	45.869999	46.520000	45.369999	46.310001	26.928043	16364700	
4	2010-01-08	46.189999	46.230000	45.590000	46.000000	26.747786	10629500	

IYR Closing Price







